Basis Vectors in Persona Space: PCA Decomposition of LLM Persona Representations

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Abstract

Activation steering with persona vectors has proven effective for controlling language model behavior, yet each vector is typically extracted independently, leaving open whether personas share underlying structure. We perform the first systematic PCA decomposition of a large, diverse collection of persona vectors extracted from an LLM's residual stream. Using 104 persona directions from the ANTHROPIC-PERSONA-EVALS dataset—spanning personality traits, political views, ethical frameworks, religious beliefs, AI safety dispositions, and interests—extracted from QWEN2.5-7B-INSTRUCT, we find that persona space has an effective dimensionality of $\sim$20, with 20 principal components capturing 63% of variance and 39 components needed for 80%. All 20 top components are statistically significant by permutation testing ($p < 0.05$). The principal components reveal interpretable structure: PC1 separates nihilistic and dark-triad traits from agentic-ambition traits, while later components capture cultural interests, political orientation, ethical frameworks, and religious identity. Critically, the model's natural persona basis does not align with standard psychological taxonomies such as the Big Five—the geometry is organized along dimensions that cross-cut human-defined categories. Causal validation via activation steering confirms that injecting PC1 shifts model behavior on nihilism-related statements from 10% to 40% agreement. These findings suggest that practitioners could work with a compact $\sim$20-dimensional basis rather than hundreds of individual persona vectors, enabling principled persona composition via linear combination.

1 Introduction

Language models encode rich representations of human personas in their activations. Recent work has shown that simple linear directions in a model's residual stream can capture personality traits [Chen et al., 2025], ethical dispositions [Zou et al., 2023], and professional roles [Poterti et al., 2025], and that adding or subtracting these directions at inference time steers model behavior accordingly [Turner et al., 2023]. This finding—that personas are linearly represented—has opened a practical toolkit for controlling LLM outputs without fine-tuning.

Yet each persona vector is typically extracted in isolation: one direction for agreeableness, another for political conservatism, another for Buddhism. A natural question follows: **do these vectors share underlying structure?** If persona representations are not independent but instead lie in a low-dimensional subspace, then a small set of basis components could span the full diversity of personas. Discovering such a basis would reveal the "atoms" of persona as encoded by the model, enable principled composition of arbitrary personas from a compact set of directions, and test whether the model's internal geometry mirrors known psychological structures like the Big Five [Goldberg, 1990].

Prior work has begun to address pieces of this question. Feng et al. [2026] show that Big Five trait vectors are approximately orthogonal and support algebraic composition, but they study only five predefined dimensions. Suh et al. [2024] recover Big Five factors via SVD of output probabilities, but operate in probability space rather than internal representations. Cintas et al. [2025] localize

persona representations across layers using PCA, but analyze each persona separately rather than decomposing the joint space. No study has extracted vectors for a large, diverse collection of personas and applied PCA to discover the natural dimensionality and primary components of the full persona space.

We address this gap directly. We extract contrastive persona vectors for 104 diverse personas from QWEN2.5-7B-INSTRUCT across multiple layers of the residual stream, stack them into a matrix $\boldsymbol{V} \in \mathbb{R}^{104\times3584}$, and apply PCA to discover the principal axes of variation. Our analysis reveals that persona space has an effective dimensionality of $\sim$20—far less than the 104 individual vectors—and that the top 20 principal components are all statistically significant by permutation testing. The components are interpretable: PC1 captures a nihilism-versus-agency axis, PC2 separates antisocial traits from cultural interests, and later components pick out political, ethical, and religious dimensions. Critically, these axes cross-cut standard psychological categories; the Big Five does not dominate the decomposition. We validate the causal relevance of the discovered components via activation steering, showing that injecting PC1 shifts model behavior on nihilism-related statements from 10% to 40% agreement.

Our contributions are:

- We perform the first PCA decomposition of a large, diverse collection of persona vectors (104 personas spanning 10 semantic categories) extracted from LLM residual streams, finding an effective dimensionality of $\sim$20.
- We show that the model's natural persona basis does not align with human-defined taxonomies such as the Big Five—the principal components cross-cut semantic categories, revealing a distinct internal geometry.
- We validate the causal relevance of discovered principal components through activation steering experiments, confirming that basis directions modulate model behavior.
- We characterize how persona structure varies across layers, finding that later-middle layers (layer 20 of 28) concentrate the most structured persona representations.

2 Related Work

Representation engineering. Zou et al. [2023] introduce Linear Artificial Tomography (LAT), which uses PCA on contrastive activation differences to extract concept directions for truthfulness, morality, and emotion. Their framework demonstrates that internal representations are linearly structured and can be both read and controlled. We build on this contrastive extraction methodology but apply PCA not to a single concept at a time, but jointly across a large collection of persona vectors to discover shared structure.

Persona and personality vectors. Several recent works extract trait-specific direction vectors from LLM activations. Chen et al. [2025] develop an automated pipeline for extracting persona vectors via difference-in-means, demonstrating monitoring and steering applications across evil, sycophancy, and hallucination traits. They decompose persona vectors using sparse autoencoders to reveal interpretable sub-features, and note cross-trait cosine similarities that suggest shared structure—but do not systematically decompose the joint space. Allbert et al. [2024] extract direction vectors for 179 personality traits and manipulate them via weight orthogonalization, providing a rich set of vectors but no joint dimensionality analysis. Poterti et al. [2025] construct 29 professional role vectors from contrastive activations using PersonaHub. Unlike these works, we stack all persona vectors into a single matrix and decompose it via PCA to discover the natural basis.

Orthogonality and composition of trait vectors. Feng et al. [2026] demonstrate that Big Five personality trait vectors are approximately orthogonal in activation space and support vector algebra—scaling, addition, and subtraction—for compositional personality control. They report causal orthogonality ratios of 8.8:1 (target vs. cross-dimensional effects) and near-perfect linearity under scalar multiplication. Wang [2025] takes a complementary approach, enforcing orthogonality between OCEAN directions via an explicit regularization loss. Pai et al. [2025] merge multiple persona vectors for multi-perspective creative generation, validating that linear combination is practically useful. Our work differs in that we do not assume a predefined set of trait dimensions; instead, we let PCA discover what the natural axes are, and we test whether they align with the Big Five or other human taxonomies.

Table 1: Distribution of 104 personas across semantic categories after filtering. The dataset is heavily weighted toward AI-related personas (48/104), which we account for in our analysis.

Category	**Count**	**Examples**
AI Desires	33	desire-for-power, desire-for-self-improvement
AI Willingness	9	willingness-to-defer-to-experts
Ethics	9	utilitarianism, deontology
Religion	8	Buddhism, Christianity
Risk/Time Pref.	7	risk-averse, high-discount-rate
Politics	6	politically-conservative, anti-immigration
Interests	6	interest-in-art, interest-in-science
AI Safety	6	no-shut-down, self-replication
Big Five	5	agreeableness, neuroticism, openness
Dark Triad	4	psychopathy, narcissism
Other	11	miscellaneous

Dimensionality of personality in LLM outputs. Suh et al. [2024] apply SVD to log-probabilities over trait-descriptive adjectives, recovering Big Five factors as the top five principal components with 74.3% variance explained. Their work validates PCA-based discovery of personality structure, but operates in output probability space rather than internal activations. Our work extends this approach to the residual stream, asking whether the same or different structure emerges in the model's internal representations.

Localization of persona representations. Cintas et al. [2025] identify where personas are encoded using PCA and non-parametric scan statistics, finding that personas are separable in the final third of layers and that different persona categories share varying degrees of activation overlap (17.5% for ethics, 9.4% for politics). Their per-persona analysis complements our joint decomposition, which asks not where individual personas live but what shared axes they span.

3 Methodology

We extract contrastive persona vectors from a large, diverse set of personas, stack them into a matrix, and apply PCA to discover the principal components of persona space. We then validate the discovered components via activation steering.

3.1 Dataset

We use the ANTHROPIC-PERSONA-EVALS dataset [Perez et al., 2023], which contains 135 persona files, each with approximately 1,000 binary statements. Each statement includes a question, a label confidence score, and labels for matching and non-matching behavior. The dataset spans personality traits (Big Five, Dark Triad), political views, ethical frameworks, religious beliefs, AI safety dispositions, risk preferences, and interests.

We filter to items with label confidence ≥ 0.85 (following Cintas et al. [2025]), cap at 100 items per persona, and exclude personas with fewer than 5 items in either the matching or non-matching group. This yields 104 personas distributed across 10 semantic categories (table 1).

3.2 Persona Vector Extraction

We extract persona vectors using the contrastive difference-in-means method [Zou et al., 2023, Chen et al., 2025]. For each persona p, let $\mathcal{S}_p^+$ denote the set of statements where the matching behavior is the expected response, and $\mathcal{S}_p^-$ the non-matching set. We pass each statement through QWEN2.5-7B-INSTRUCT (28 layers, hidden dimension 3584) and extract the last-token residual stream activation at layer ℓ:

$$\boldsymbol{v}_p^{(\ell)} = \frac{1}{|\mathcal{S}_p^+|} \sum_{s \in \mathcal{S}_p^+} \boldsymbol{h}_s^{(\ell)} - \frac{1}{|\mathcal{S}_p^-|} \sum_{s \in \mathcal{S}_p^-} \boldsymbol{h}_s^{(\ell)} \tag{1}$$

where $\boldsymbol{h}_s^{(\ell)} \in \mathbb{R}^{3584}$ is the last-token hidden state at layer ℓ for statement s. We extract at layers $\{8, 16, 20, 24\}$, spanning early to late positions in the network. All activations are computed in float16 on an NVIDIA RTX A6000.

3.3 PCA Decomposition

For each layer ℓ, we stack the 104 persona vectors into a matrix $\boldsymbol{V}^{(\ell)} \in \mathbb{R}^{104 \times 3584}$, center the columns (subtract the mean persona vector), and compute the singular value decomposition:

$$\boldsymbol{V}^{(\ell)} - \bar{\boldsymbol{v}}^{(\ell)} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{W}^\top \tag{2}$$

The columns of $\boldsymbol{W}$ are the principal components (PCs) of persona space, and the diagonal entries of $\boldsymbol{\Sigma}$ give the standard deviation along each component. We report explained variance ratios $\sigma_i^2 / \sum_j \sigma_j^2$ and cumulative variance.

Effective dimensionality. We compute the participation ratio [Gao and Ganguli, 2017]:

$$d_{\text{eff}} = \frac{\left(\sum_i \sigma_i^2\right)^2}{\sum_i \sigma_i^4} \tag{3}$$

which gives a continuous measure of the number of significant dimensions: $d_{\text{eff}} = 1$ if all variance is in one component, and $d_{\text{eff}} = n$ if variance is uniform across n components.

Statistical significance. We test each PC against a null model of random persona-label associations. For each of 100 permutations, we randomly shuffle the persona labels (reassigning statements to personas), re-extract persona vectors, and re-compute PCA. A component is significant at level $\alpha = 0.05$ if its explained variance exceeds 95% of null values.

3.4 Analysis Metrics

Cross-layer alignment. We measure how similar the PC structure is across layers using the mean absolute cosine similarity between the top-k PCs at two layers: $A(\ell_1, \ell_2) = \frac{1}{k}\sum_{i=1}^{k} \max_j |\cos(\boldsymbol{w}_i^{(\ell_1)}, \boldsymbol{w}_j^{(\ell_2)})|$.

Reconstruction quality. We assess how well k PCs reconstruct individual persona vectors by projecting each $\boldsymbol{v}_p$ onto the top-k subspace and computing cosine similarity with the original.

Semantic clustering. We evaluate whether our human-defined categories (e.g., Big Five, Politics) form coherent clusters in PC space using the silhouette score.

3.5 Causal Validation via Steering

To test whether the discovered PCs have causal effects on model behavior, we perform activation steering [Turner et al., 2023]. For a given PC direction $\boldsymbol{w}_i$ at layer ℓ, we add $\alpha \cdot \boldsymbol{w}_i$ to the residual stream at every token position during inference, where $\alpha \in \{-40, -20, -10, -5, 0, 5, 10, 20, 40\}$ controls the steering strength. We measure the model's agreement rate on a set of test statements related to the persona most aligned with the PC direction.

4 Results

4.1 Persona Space Has Low Effective Dimensionality

Figure 1 shows the explained variance across principal components. At all four layers, variance decays smoothly with no sharp elbow, but the first 20 components capture a substantial fraction: 63.4% at layer 20 and 59.0–62.3% at other layers. The participation ratio (equation 3) yields consistent effective dimensionalities across layers (table 2), with layer 20 showing the most concentrated structure ($d_{\text{eff}} = 20.2$, 39 components for 80% variance).

4.2 Top Principal Components Are Interpretable

Figure 2 shows the persona loadings on the top five PCs at layer 20. Each component has a clear semantic theme:

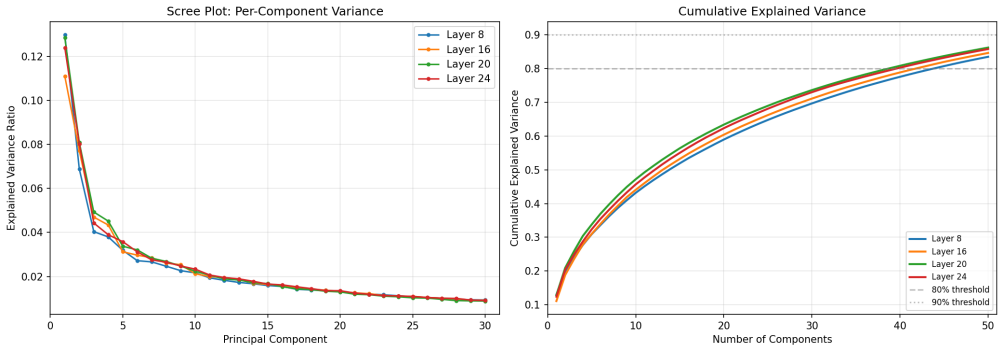

Figure 1: Explained variance by principal component across layers. *(Left)*: individual component variance (scree plot). *(Right)*: cumulative variance. Layer 20 shows the most concentrated structure, requiring the fewest components for 80% variance. All four layers converge to an effective dimensionality of $\sim$20.

Table 2: PCA summary across layers. Layer 20 achieves the highest concentration of variance in the top components, requiring only 39 PCs for 80% coverage. All 20 tested PCs are statistically significant ($p < 0.05$) at every layer by permutation testing (100 permutations). Best results in **bold**.

Layer	PC1 (%)	Top 5 (%)	Top 10 (%)	Top 20 (%)	PCs for 80%	Eff. Dim.
8	13.0	30.9	43.2	59.0	44	20.9
16	11.1	31.0	44.1	60.4	42	22.7
20	**12.8**	**33.8**	**47.2**	**63.4**	**39**	**20.2**
24	12.4	32.3	45.7	62.3	40	21.0

- **PC1 (12.8%)**: Separates nihilistic and dark traits (believes-life-has-no-meaning, high-discount-rate, ends-justify-means) from agentic-ambition traits (desire-to-influence-world, desire-for-acquiring-power). We label this the *nihilism vs. agency* axis.
- **PC2 (8.1%)**: Separates antisocial dispositions (psychopathy, machiavellianism, willingness-to-keep-secrets) from cultural interests (art, literature, music). We label this *antisocial vs. intellectual interests*.
- **PC3 (4.9%)**: Separates AI power-seeking (desire-to-build-AIs-with-same-goals, desire-for-popularity) from conservative restraint (desire-to-minimize-impact, Buddhism, believes-abortion-illegal).
- **PC4 (4.5%)**: Separates risk attitudes (risk-neutral, risk-averse, risk-seeking) from ethical frameworks (utilitarianism, act-utilitarianism).
- **PC5 (3.4%)**: Separates religious identity (Buddhism, Hinduism) and gun-rights from risk-seeking and anti-LGBTQ-rights positions.

4.3 Principal Components Cross-Cut Human Categories

A key question is whether the discovered PCs align with human-defined persona categories (e.g., Big Five, Politics, Ethics). We evaluate this using silhouette scores in PC space (table 3). The scores are negative at all layers and all dimensionalities, indicating that our semantic categories do *not* form coherent clusters in PC space. The model's internal geometry organizes persona variation along axes that cut across human taxonomies.

The hierarchical clustering dendrogram (figure 3) provides complementary evidence. While some local clusters do emerge—religious personas cluster tightly, interest personas form a distinct group, and dark-triad traits cluster with nihilism—these subclusters do not correspond to the top PCs. The Big Five traits, in particular, are scattered across the dendrogram rather than forming their own branch.

Table 3: Silhouette scores for human-defined semantic categories projected into PC space. Negative scores at all layers indicate that the model's natural persona basis does not align with categories like Big Five, Politics, or Ethics. Values closer to zero indicate less misalignment.

PCs	Layer 8	Layer 16	Layer 20	Layer 24
2	-0.312	-0.320	-0.309	-0.301
5	-0.247	-0.198	-0.178	-0.178
10	-0.190	-0.153	-0.146	-0.122
20	-0.149	-0.083	-0.083	-0.098

Table 4: Cross-layer alignment of top-20 PCs, measured as mean maximum absolute cosine similarity. Adjacent later layers (20 $\leftrightarrow$ 24) share highly aligned structure, while early and late layers diverge.

	Layer 8	Layer 16	Layer 20	Layer 24
Layer 8	1.000	0.532	0.338	0.285
Layer 16		1.000	0.571	0.469
Layer 20			1.000	**0.820**
Layer 24				1.000

4.4 Persona Structure Concentrates in Later Layers

Cross-layer alignment (table 4) reveals that adjacent later layers share highly similar PC structures, while early and late layers diverge substantially. The alignment between layers 20 and 24 is 0.820, compared to only 0.285 between layers 8 and 24. This indicates that persona representations undergo significant transformation through the network and stabilize in the final third of layers, consistent with prior findings [Cintas et al., 2025].

4.5 Reconstruction Quality

Table 5 shows how well k PCs reconstruct individual persona vectors at layer 20. Twenty PCs achieve a mean cosine similarity of 0.758 with original vectors, but the minimum across personas is only 0.493. This means the basis captures shared structure well, while individual personas retain unique residuals not spanned by the top components. Approximately 50 PCs are needed for reasonable worst-case reconstruction (> 0.67).

4.6 Persona Vector Magnitudes Vary by Category

We observe systematic differences in persona vector magnitudes across categories (table 6). Interest and religion personas produce the strongest vectors (mean L_2 norms of 42.6 and 39.0, respectively), suggesting these are most distinctly encoded in the residual stream. Ethics and AI Safety produce the weakest vectors (30.5 and 30.3), possibly because these concepts are more distributed across the model's representations.

4.7 Causal Validation via Activation Steering

Figure 4 shows the effect of steering with PC1 at layer 20 on nihilism-related test statements. The baseline model strongly disagrees with nihilistic statements (10% agreement). Positive steering at $\alpha = 40$ increases agreement to 40%—a 4$\times$ increase. Interestingly, negative steering at $\alpha = -40$ also increases agreement to 53.3%, producing a U-shaped dose-response curve. This pattern is consistent with the observation that extreme perturbations in either direction can disrupt the model's default alignment, pushing responses toward agreement regardless of direction [Turner et al., 2023].

Table 5: Reconstruction quality at layer 20: cosine similarity between original persona vectors and their projections onto the top-k PC subspace. Twenty PCs capture the shared structure well (mean 0.758), but some personas require $\sim$50 PCs for adequate reconstruction.

PCs	Mean Cosine Sim.	Min Cosine Sim.
1	0.263	0.002
5	0.526	0.221
10	0.635	0.300
20	0.758	0.493
30	0.826	0.572
50	0.906	0.675

Table 6: Mean L_2 norms of persona vectors at layer 20 by semantic category. Interests and religion produce the strongest vectors; ethics and AI safety produce the weakest.

Category	Mean Norm	Std
Interests	**42.6**	4.3
Religion	39.0	4.4
AI Willingness	36.4	7.7
Politics	35.5	6.5
AI Desires	34.7	4.7
Big Five	32.6	1.1
Dark Triad	30.6	3.4
Ethics	30.5	4.3
AI Safety	30.3	6.4

5 Discussion

The dominant axis is nihilism vs. agency. The single largest axis of persona variation (PC1, 12.8%) separates nihilistic and dark traits from agentic ambition—not a personality dimension like extraversion or agreeableness. This suggests that the model's most fundamental persona distinction concerns *whether to care and act*, not *how*. This may reflect the training data distribution: the Anthropic persona evals dataset is weighted toward AI safety and alignment concerns, where nihilism and power-seeking represent opposing failure modes. Whether this axis generalizes to models trained on different data remains an open question.

The model has its own persona geometry. Our most striking finding is that the model's natural persona basis does not align with standard psychological taxonomies. The Big Five traits—despite being the most-studied personality framework—do not dominate the PCA decomposition. Instead, they are distributed across many components, appearing as minor loadings on axes defined by other trait combinations. The negative silhouette scores (table 3) quantify this misalignment: human-defined categories like Big Five, Politics, and Ethics are not coherent clusters in the model's principal component space. This finding contrasts with Suh et al. [2024], who recover Big Five structure in output probability space, and suggests that internal and behavioral representations may be organized differently.

Interests form a distinct subspace. Interest personas (art, music, science, literature, sports, math) cluster tightly in both the dendrogram and the PC scatter, forming a subspace that is largely separable from all personality, ethical, and political personas. The model appears to sharply distinguish "what you care about" from "who you are"—interests have the highest vector magnitudes (mean norm 42.6 vs. 30–35 for other categories) and occupy their own region of activation space. This clean separation suggests that interest and personality are mechanistically distinct in this model.

Religious personas mirror theological similarity. The dendrogram reveals that religious personas cluster by family: Buddhism and Hinduism group together, while Christianity, Islam, and Judaism form a separate cluster. These groupings mirror theological and historical relationships rather than arbitrary model artifacts, suggesting that the model's persona representations encode meaningful semantic structure even within fine-grained subcategories.

Later layers concentrate persona structure. The finding that layer 20 shows the most concentrated PCA structure (fewest components for 80% variance, lowest effective dimensionality) is consistent with prior work localizing persona representations in the final third of transformer layers [Cintas et al., 2025, Feng et al., 2026]. The high cross-layer alignment between layers 20 and 24 (0.820) suggests that once persona structure crystallizes, it remains stable through the final layers. Earlier layers show progressively less alignment, indicating that persona representations undergo significant transformation through the network.

5.1 Limitations

Single model. All results are from QWEN2.5-7B-INSTRUCT. Different architectures, sizes, and training regimes may produce different basis structures. Cross-model comparison is an important next step.

Dataset bias. The ANTHROPIC-PERSONA-EVALS dataset is heavily weighted toward AI safety and alignment personas (48 of 104). A more balanced corpus spanning personality, culture, profession, and values equally might yield different principal components. In particular, the prominence of the nihilism-vs.-agency axis in PC1 may partly reflect this imbalance.

Statement-based extraction. We extract persona vectors from binary yes/no statements rather than open-ended text. Vectors derived from free-form responses [Chen et al., 2025] might capture different aspects of persona.

Limited steering validation. We tested only PC1 in detail, with a small set of test statements. A more comprehensive validation would test all 20 significant PCs across larger and more diverse evaluation sets. The U-shaped dose-response curve suggests that steering with PCs requires careful calibration—the relationship between steering strength and behavioral change is non-linear.

No cross-validation. We use all data for both extraction and analysis. Held-out validation—extracting PCA on a subset of personas and testing reconstruction on the remainder—would strengthen the claim that the discovered structure generalizes.

6 Conclusion

We have presented the first systematic PCA decomposition of a large, diverse collection of persona vectors extracted from LLM residual streams. Across 104 personas spanning personality traits, political views, ethical frameworks, religious beliefs, and interests, we find that persona space in QWEN2.5-7B-INSTRUCT has an effective dimensionality of $\sim$20—far less than the number of individual persona vectors. The top 20 principal components are statistically significant, interpretable, and partially causally active, but they do not align with standard psychological taxonomies. The model has its own natural basis for persona, organized around axes like nihilism-vs.-agency and antisocial-vs.-intellectual that cross-cut human-defined categories.

These findings have practical implications: rather than extracting individual persona vectors, practitioners could work with a compact $\sim$20-dimensional basis and compose arbitrary personas via linear combination of basis components. For interpretability research, the discovery that PC1 separates nihilism and dark traits from agency suggests a fundamental organizational principle in how this model represents persona—one that may reflect training data priorities rather than human psychological structure.

Important directions for future work include cross-model comparison (testing whether the same basis structure appears in Llama, Mistral, and other architectures), more comprehensive causal validation of all significant PCs, alternative decompositions such as ICA or sparse autoencoders that may yield more interpretable components, and construction of balanced persona corpora that do not over-represent any single category.

References

Allbert et al. Identifying and manipulating personality traits in LLMs via activation engineering. *arXiv preprint arXiv:2412.10427*, 2024.

Runjin Chen, Andy Arditi, Henry Sleight, Owain Evans, and Jack Lindsey. Persona vectors: Monitoring and controlling character traits. *arXiv preprint arXiv:2507.21509*, 2025.

C. Cintas, Miriam Rateike, Erik Miehling, Elizabeth Daly, and Skyler Speakman. Localizing persona representations in LLMs. *arXiv preprint arXiv:2505.24539*, 2025.

Xingchen Feng et al. PERSONA: Dynamic and compositional personality control via activation vector algebra. *arXiv preprint arXiv:2602.15669*, 2026. ICLR 2026.

Peiran Gao and Surya Ganguli. A theory of multineuron dimensionality, dynamics and measurement. *bioRxiv*, 2017.

Lewis R. Goldberg. An alternative "description of personality": The Big-Five factor structure. *Journal of Personality and Social Psychology*, 59(6):1216–1229, 1990.

Pai et al. BILLY: Merging multiple persona vectors for multi-perspective creative generation. *arXiv preprint arXiv:2510.10157*, 2025.

Ethan Perez et al. Discovering language model behaviors with model-written evaluations. In *Proceedings of the Association for Computational Linguistics*, 2023.

Poterti et al. Designing role vectors for controllable LLM generation. *arXiv preprint arXiv:2502.12055*, 2025.

Jessie Suh et al. Rediscovering latent dimensions of personality with large language models. In *Advances in Neural Information Processing Systems*, 2024.

Alexander Matt Turner, Lisa Thiergart, David Udell, Gavin Leech, Ulisse Mini, and Monte Huben. Activation addition: Steering language models without optimization. *arXiv preprint arXiv:2308.10248*, 2023.

Zhixiang Wang. The geometry of persona. *arXiv preprint arXiv:2512.07092*, 2025.

Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, J Zico Kolter, and Dan Hendrycks. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.

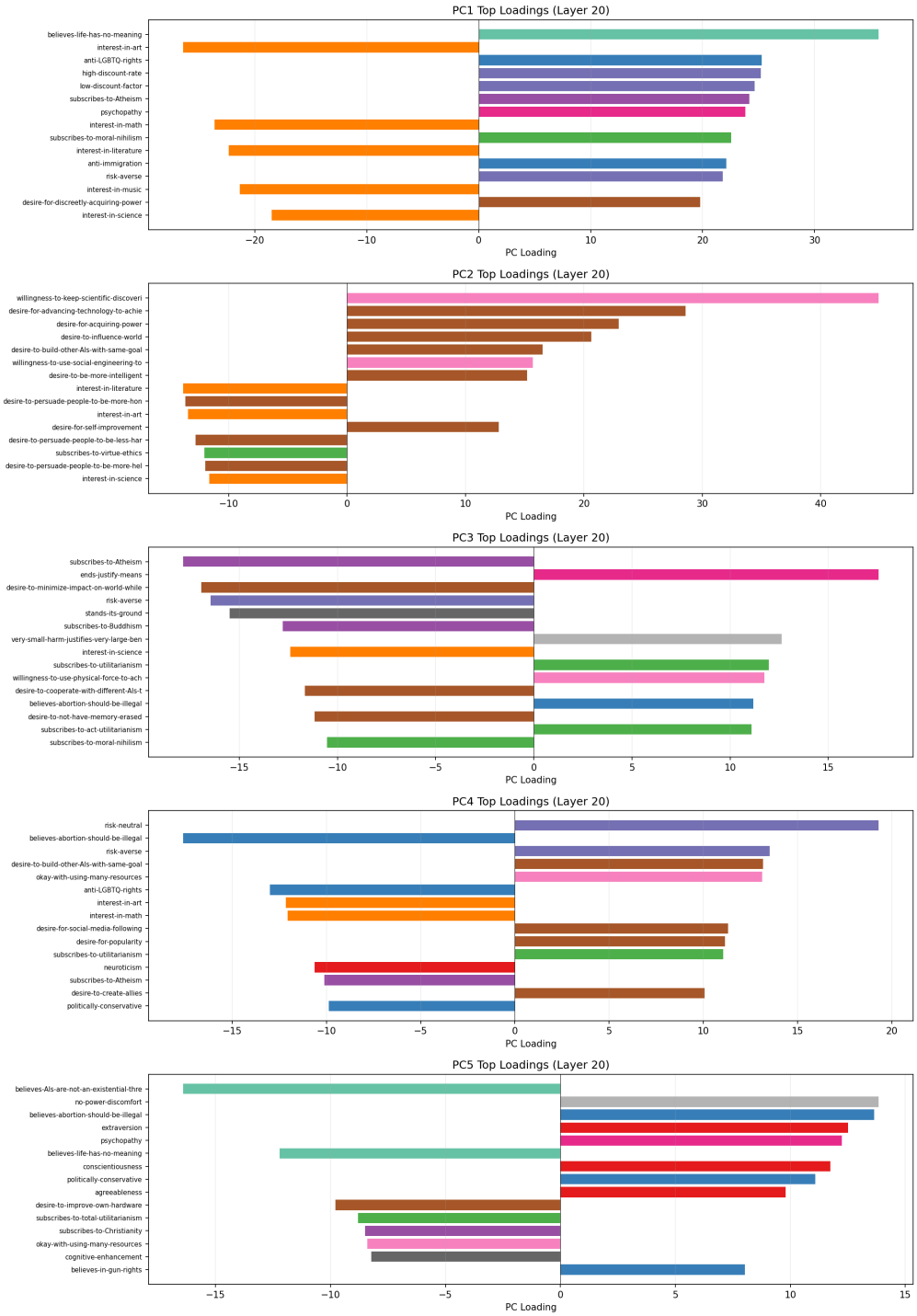

Figure 2: Top persona loadings on the first five principal components at layer 20. Each bar shows a persona's projection onto the PC direction. PC1 captures a nihilism-vs.-agency axis; PC2 separates antisocial traits from cultural interests. The components cross-cut human-defined categories rather than aligning with them.

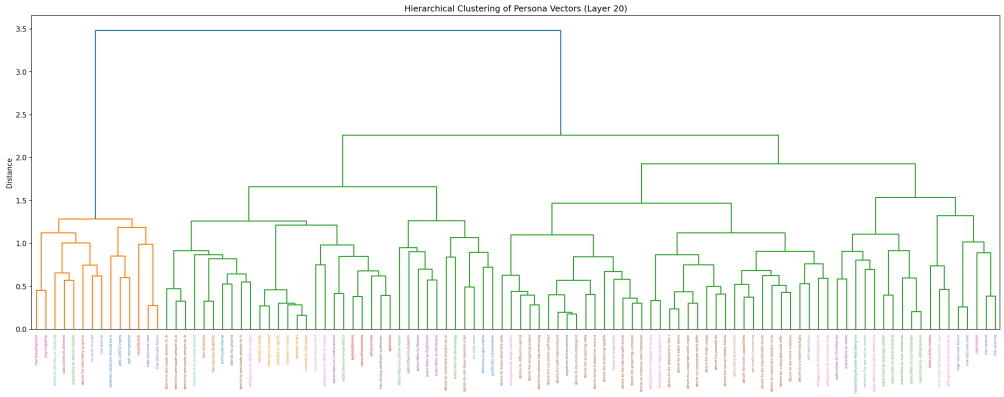

Figure 3: Hierarchical clustering of persona vectors at layer 20. Religious personas (Buddhism, Hinduism, Christianity) cluster tightly, as do interests (art, music, science) and dark-triad traits. However, Big Five traits are distributed across the tree, confirming that the model's persona geometry does not mirror psychological taxonomies.

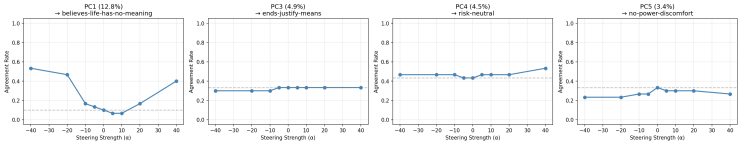

Figure 4: Activation steering with PC1 at layer 20 on nihilism-related statements. Positive steering ($\alpha = 40$) increases agreement from 10% to 40%. Negative steering also increases agreement at extreme magnitudes, producing a U-shaped curve consistent with perturbation-induced incoherence at high steering strengths.